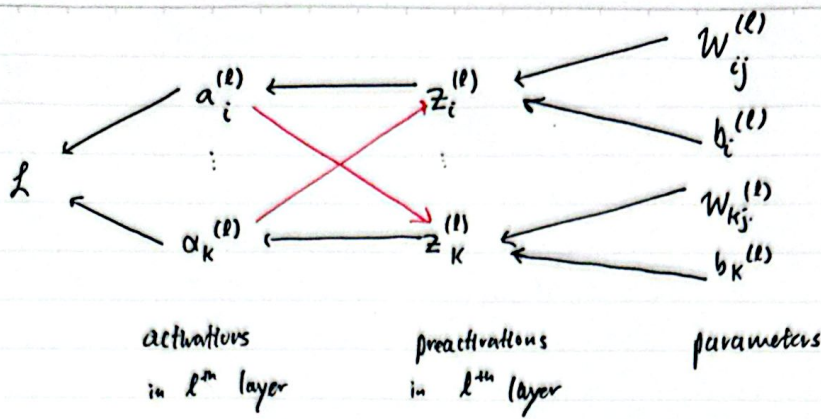
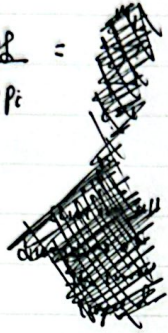




→ interdependence.

before:

$$\frac{\partial \mathcal{L}}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial a_i^{(l)}} \cdot \frac{\partial a_i^{(l)}}{\partial z_i^{(l)}} \cdot \frac{\partial z_i^{(l)}}{\partial p_i} \quad \text{where } p_i \in \{b_i^{(l)}, w_{ij}^{(l)}\}$$



this was true when our activation functions are non interdependent, i.e. $a_i^{(l)}$ does not depend on $z_k^{(l)}$ $i \neq k$.

but for interdependent activation function, $a_k^{(l)}$ depends on $z_i^{(l)}$ (see diagram)

thus:

$$\frac{\partial \mathcal{L}}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial a_i^{(l)}} \cdot \frac{\partial a_i^{(l)}}{\partial z_i^{(l)}} \cdot \frac{\partial z_i^{(l)}}{\partial p_i}$$

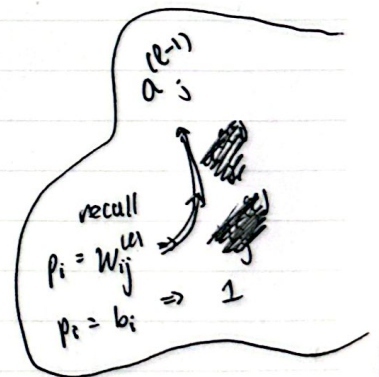
$$+ \frac{\partial \mathcal{L}}{\partial a_k^{(l)}} \cdot \frac{\partial a_k^{(l)}}{\partial z_i^{(l)}} \cdot \frac{\partial z_i^{(l)}}{\partial p_i}$$

↳ previously 0 if not interdependent

⇒ generalize:

$$\frac{\partial \mathcal{L}}{\partial p_i} = \left(\sum_{c=1}^{m_L} \frac{\partial \mathcal{L}}{\partial a_c^{(l)}} \cdot \frac{\partial a_c^{(l)}}{\partial z_i^{(l)}} \right) \cdot \frac{\partial z_i^{(l)}}{\partial p_i}$$

sum over all layer ~~width~~ dimensions



implement:
network.cpp

curr_gradient still stores

$$\left[\frac{\partial \mathcal{L}}{\partial a_c^{(l)}} \right]_{c=1}^{m_L}$$

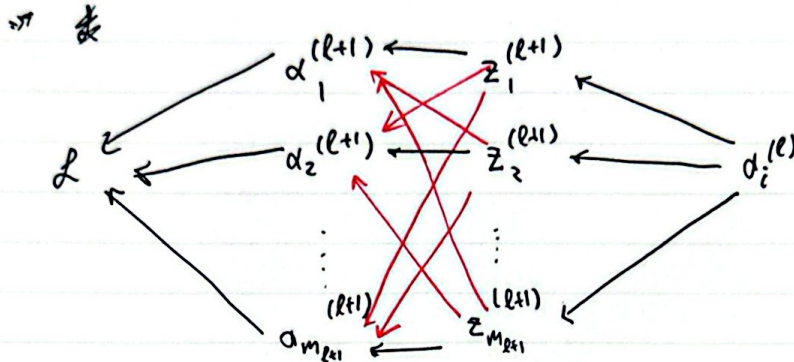
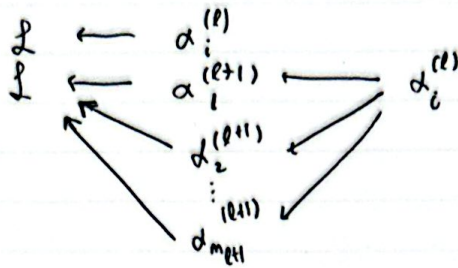
as result.

given i

immed_value $\approx$ sum $\{$
 $c=0 \rightarrow m_L-1$
 change!

curr_gradient [c] * jacobian ($z^{(l)}, c, i$) }

do we need to update formula for $\frac{\partial \mathcal{L}}{\partial v_i^{(l)}}$



before: $\frac{\partial \mathcal{L}}{\partial \alpha_i^{(l)}} = \sum_{k=1}^{m_{l+1}} \frac{\partial \mathcal{L}}{\partial \alpha_k^{(l+1)}} \cdot \frac{\partial \alpha_k^{(l+1)}}{\partial z_k^{(l+1)}} \cdot \frac{\partial z_k^{(l+1)}}{\partial \alpha_i^{(l)}}$

yes... have to update formula...

now: $\frac{\partial \mathcal{L}}{\partial \alpha_i^{(l)}} = \sum_{k=1}^{m_{l+1}} \sum_{c=1}^{m_{l+1}} \frac{\partial \mathcal{L}}{\partial \alpha_k^{(l+1)}} \cdot \frac{\partial \alpha_k^{(l+1)}}{\partial z_c^{(l+1)}} \cdot \frac{\partial z_c^{(l+1)}}{\partial \alpha_i^{(l)}}$

$\frac{\partial \mathcal{L}}{\partial \alpha_i^{(l)}} = \sum_{k=1}^{m_{l+1}} \left(\frac{\partial \mathcal{L}}{\partial \alpha_k^{(l+1)}} \cdot \sum_{c=1}^{m_{l+1}} \frac{\partial \alpha_k^{(l+1)}}{\partial z_c^{(l+1)}} \cdot w_{ci}^{(l+1)} \right)$

double sum

$i = 0, \dots, m_l$

implemented...?

C in code implm.

inner loop on the `z` of network.cpp

gradient...